

Anderson Morgan

Senior Data Engineer

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SUMMARY

Senior Data Engineer with **9+ years** of experience architecting large-scale, cloud-native data platforms across **AWS**, and **Azure**. Skilled in building high-throughput **streaming**, **lakehouse**, and **ELT** pipelines using **Spark**, **EMR**, **Glue**, **Kafka**, **Kinesis**, **Flink**, **dbt**, and **Airflow** to power real-time analytics in **fintech**, **healthcare**, and **e-commerce**. Deep expertise in **Delta Lake**, **Iceberg**, **Snowflake**, **Redshift**, and modern observability stacks including **Datadog**, **Prometheus**, and **ELK**. Known for driving modernization, implementing regulatory-grade governance with Great Expectations, and delivering scalable, secure, high-performance data systems.

TECHNICAL SKILLS

- **Programming:** Python, Scala, Java, SQL
- **Data Processing & ETL:** Apache Spark, Apache Airflow, dbt, Apache Beam, ETL/ELT pipeline development
- **Cloud Platforms:** AWS (S3, Glue, EMR, Lambda, Athena, Redshift, Kinesis), Azure (Data Factory, Synapse, Blob Storage, Event Hubs)
- **Data Warehousing:** Snowflake, Amazon Redshift, Dimensional modeling, ELT patterns, performance tuning
- **Data Lakes & Storage:** Amazon S3, Azure Data Lake Storage, Delta Lake, Apache Iceberg
- **Streaming & Real-Time:** Apache Kafka, Apache Flink, Event-driven pipelines, real-time ingestion
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, CI/CD pipelines, version control
- **Monitoring & Observability:** Prometheus, Grafana, Datadog, ELK Stack, Logging, metrics
- **Data Governance & Quality:** Great Expectations, Apache Atlas, Data contracts, schema validation
- **BI & Visualization:** Tableau, Looker, Power BI, Dashboarding, semantic modeling, SQL analytics
- **Enterprise Delivery, Leadership & Collaboration:** Agile, Scrum/Kanban, JIRA

PROFESSIONAL EXPERIENCE

1. Senior Data Engineer | CGI | Los Angeles, CA | 08/2023 – Present

- Architected and delivered AWS-native, large-scale data pipelines using **Spark** on **EMR**, **Glue**, **Lambda**, **Step Functions**, and **Kinesis** to power real-time **financial** and transactional processing.
- Led high-throughput streaming ingestion frameworks leveraging **Kafka** and **Kinesis** to handle millions of payment events with sub-second latency for fraud detection and order lifecycle analytics.
- Designed and implemented **Lakehouse architecture** on **S3** with **Delta Lake/Iceberg**, enabling ACID transactions, schema evolution, and unified batch/stream processing.
- Developed **dbt-based ELT models** supporting merchant analytics, customer segmentation, revenue forecasting, and marketplace intelligence across **fintech** and **e-commerce** domains.

- Modernized legacy **ETL** into **distributed Spark pipelines**, cutting processing time by 60%+ and reducing compute cost through EMR autoscaling and spot-instance optimization.
- Engineered **Airflow DAGs** orchestrating cross-domain data flows, SLAs, and dependency management for financial reporting, settlements, and operational analytics.
- Designed and optimized **Snowflake** and **Redshift** warehouse models, applying dimensional modeling, workload isolation, and performance tuning for high-concurrency analytics.
- Built a metadata-driven ingestion framework using **Python**, **Jinja2**, and **YAML**, enabling rapid onboarding of new financial APIs, payment processors, and partner integrations with minimal engineering effort
- Built real-time fraud scoring pipelines integrating **ML features**, streaming enrichment, and event-driven microservices to strengthen risk and compliance operations.
- Established data quality and governance frameworks using **Great Expectations** and **schema contracts**, ensuring regulatory-grade accuracy for payments, refunds, and compliance datasets.
- Implemented secure, compliant data flows aligned with **PCI** and **SOC2**, including encryption, **IAM/RBAC**, tokenization, and full auditability.
- Designed and optimized **PySpark** and **Databricks** pipelines to process large-scale transactional and behavioral datasets, enabling faster analytics and reducing compute cost across **fintech** and **e-commerce** workloads.
- Built automated ingestion and transformation workflows using **Fivetran**, and **SQL**, delivering reliable **ELT pipelines** and accelerating data availability for product, finance, and risk teams
- Built interactive **Looker** dashboards and semantic models that enabled product, **finance**, and operations teams to explore KPIs, customer behavior, and real-time business performance.
- Delivered high-quality ad-hoc data outputs by writing optimized SQL and producing clean, audit-ready **CSV/Excel** extracts for finance, operations, compliance, and executive stakeholders.
- Developed observability dashboards with **Datadog**, **Prometheus**, and **ELK** to monitor pipeline health, latency, throughput, and cost anomalies across distributed systems.
- Implemented unified data governance and quality checks across **Databricks** pipelines, ensuring consistent schemas, validated metrics, and trusted data for BI and **machine-learning** use cases.
- Collaborated with **product**, **engineering**, and **data science teams** to deliver analytics capabilities for checkout optimization, pricing intelligence, and customer behavior insights.
- Led migration of **e-commerce** data workloads to **AWS serverless** and **containerized architectures** using **Lambda**, **ECS**, and **Terraform IaC**.
- Mentored engineers on distributed data patterns, AWS best practices, and scalable pipeline design, elevating team delivery velocity and architectural maturity.

2. Data Engineer | GE Healthcare | Chicago, IL | 05/2020 – 07/2023

- Engineered large-scale **ETL/ELT pipelines** using **Apache Spark**, **Python**, and **Scala**, reducing claims-processing latency by 40% across multiple healthcare data products.
- Designed and deployed **HIPAA-compliant** ingestion frameworks on **Azure Data Factory**, **Azure Synapse**, and **Azure Blob Storage**, enabling secure, automated processing of PHI at multi-TB scale.
- Built end-to-end orchestration workflows with **Apache Airflow** and **dbt**, improving data reliability and cutting pipeline failures by 60%.
- Engineered a unified healthcare data lake using **Delta Lake** and **Apache Iceberg** on **Azure Data Lake Storage**, standardizing storage formats and improving schema evolution for regulatory datasets.
- Implemented data governance using **Great Expectations** and **Apache Atlas** to validate data quality and track lineage across healthcare datasets.

- Developed real-time eligibility and claims-event pipelines using **Apache Kafka**, **Azure Event Hubs**, and **Apache Flink**, enabling sub-second event propagation for clinical decision systems.
- Implemented dimensional models and ELT patterns in **Snowflake** and **Azure Synapse Dedicated SQL Pools**, improving analytic query performance for actuarial and care-management teams.
- Built **Datadog** dashboards to monitor infrastructure, applications, and data pipelines, helping teams quickly identify issues and maintain system reliability.
- Built scalable data-quality frameworks using **Great Expectations** and **data contracts**, ensuring consistent validation of clinical, claims, and provider datasets.
- Developed unified **Datadog** observability dashboards aggregating **infrastructure**, **application**, and **data-pipeline** metrics, empowering engineering teams to proactively detect anomalies and prevent production-impacting incidents.
- Modeled and tuned **SQL** and **NoSQL** datasets to support high-performance reporting, real-time dashboards, and scalable analytical workloads across distributed systems.
- Developed interactive analytics experiences in **Power BI**, **Tableau**, and **Looker**, enabling business stakeholders to explore KPIs, customer behavior, and operational insights with self-service dashboards.
- Integrated **enterprise lineage** and **governance** using **Apache Atlas**, enabling full traceability of PHI transformations for audits and compliance reviews.
- Implemented end-to-end ELT workflows on **Databricks**, integrating **SQL**, **Delta Lake**, and **orchestration tools** to deliver reliable, high-quality data to downstream consumers.
- Containerized and deployed micro-batch and streaming workloads using **Docker**, **Kubernetes**, and **Terraform**, improving deployment consistency across Azure environments.
- Established **CI/CD pipelines** for data engineering workloads using Git-based workflows and automated testing, reducing release cycle time by 50%.
- Implemented observability for **healthcare pipelines** using **Prometheus**, **Grafana**, and **ELK**, enabling proactive detection of data drift and SLA risks.
- Designed **Apache Iceberg**-based **lakehouse** tables on **Azure Data Lake Storage**, enabling ACID transactions, time-travel queries, and schema evolution to support regulatory-grade healthcare data pipelines.
- Partnered with clinicians, analysts, and compliance teams to translate regulatory requirements into scalable data models and governed ingestion patterns.
- Led cross-functional **Agile** delivery, mentoring engineers and driving architectural decisions for modernization of legacy healthcare data systems.

3. Data Analytics | Pitney Bowes | Austin, TX | 07/2016 – 04/2020

- Analyzed **50M+ daily e-commerce** and **retail transactions** using outputs from **Apache Spark** and **Apache Airflow** to uncover customer behavior trends and operational bottlenecks, contributing to a **65%** reduction in reporting latency.
- Conducted deep-dive analytics on **Amazon Redshift** warehouse data to support merchandising, inventory optimization, and customer segmentation initiatives for a top-10 retail brand.
- Interpreted real-time clickstream, cart, and purchase event data generated from **Apache Kafka** and **Apache Flink** pipelines to drive insights for personalization and digital conversion strategies.
- Leveraged **Delta Lake** datasets on **Amazon S3** to analyze structured and semi-structured data from 30+ retail systems, enabling cross-functional reporting across product, supply chain, and customer domains.
- Developed automated analytical workflows using **Python** and advanced **SQL**, reducing manual reporting effort and enabling self-service analytics for business teams.

- Used **AWS Glue Data Catalog** and **Amazon Athena** to run cost-efficient exploratory analyses on petabyte-scale product and customer datasets, supporting ad-hoc business questions and strategic planning.
- Built version-controlled SQL models using **dbt** to standardize KPI definitions and improve consistency across merchandising, supply chain, and finance analytics.
- Analyzed real-time POS and inventory data from 500+ retail locations via **AWS Kinesis Data Streams** and **Kinesis Firehose** to support live inventory dashboards and improve stockout prediction accuracy.
- Implemented data quality checks using **Great Expectations** to ensure accuracy and reliability of analytical datasets used for executive reporting and operational decision-making.
- Designed and delivered executive dashboards in **Tableau** and **Power BI**, consolidating KPIs such as GMV, AOV, conversion rate, and inventory turnover for C-suite visibility.
- Collaborated with data science teams to prepare analytical features and datasets using **Apache Spark MLlib** and **Python**, supporting customer segmentation and recommendation model development.
- Partnered with engineering teams to ensure analytical datasets were reliable, timely, and aligned with business requirements, improving data availability for analysts and business users.
- Monitored analytical data freshness and SLA compliance using **ELK Stack** and **Grafana**, ensuring accuracy of downstream reporting.
- Worked cross-functionally with product, merchandising, and operations teams to translate business questions into analytical requirements and deliver actionable insights.

EDUCATION

The University of Texas at Austin | Austin, TX | 09/2012 – 06/2016

Bachelor's degree in computer science

ADDITIONAL INFORMATION

- **US Citizen - No sponsorship required**
- **Eligible for federal contracting roles**
- **Cloud platforms: AWS, Azure**
- **DevOps & CI/CD experience**
- **Agile/Scrum team leadership**